

The “Berlin Families”

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"If you spent half as much time paying attention to me as you do studying that sculpture thing, you would understand me a whole lot better."

- Sherry Thompson, 2010

"At a million-to-one against, the odds of me ever solving K4 are still a million times better than the odds of me ever understanding you."

- Randy Thompson, 3 Seconds Later (unspoken)

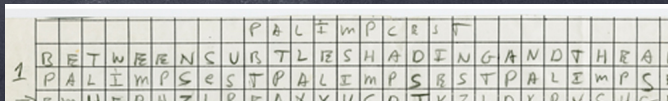
"You're right dear. I'll do better."

- Randy Thompson, 5 Seconds Later (said out loud)

K4 Breakthrough (2013)

- "... you need to solve the technique first and then go for the puzzle." – Ed Scheidt, Wired (21 Jan 2005)
 - There would have to be clues on or around Kryptos
 - Based on those clues, you need to try some methods until you find something noteworthy
- In March 2013, Scott Perry made a breakthrough...
 - Using only characters carved next to one another on the sculpture and numbers found in the plaintext, he obtained BERLIN in the right location
 - He employed methods used in K1, K2, and K3

Scott's Method (Step 1)

- Quag 3 using first 11 ciphertext chars on Kryptos
 - These characters are "EMUFPHZLRFA"
 - K2 says "they used the Earth's Magnetic Field"
 - In E&M, the Greek letter μ (pronounced "mu") is used to represent magnetic permeability. **EMUF** = **Earth's Magnetic Field**
 - K3 references "upper lefthand corner"
 - Arrow on Jim's work sheet 

[illegible]

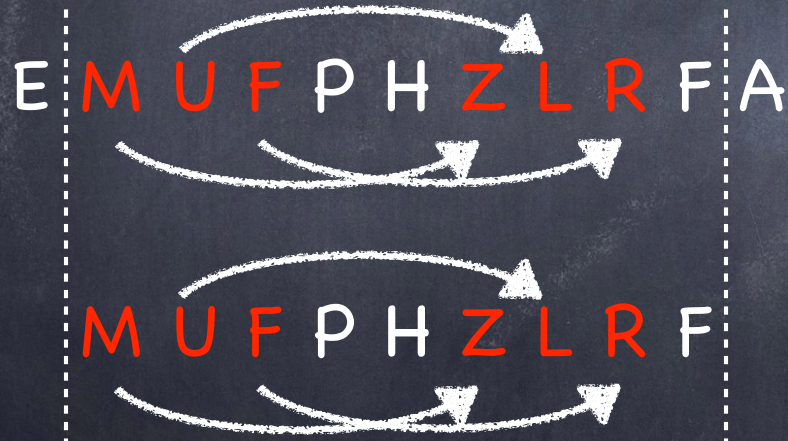
Scott's Method (Step 2)

- "Skip-decrypt" starting at 77th character and skipping every 38 chars until done
 - 38 degrees North, 77 degrees West (in K2 plaintext)
- Result:
PGZFGTEZPFCYGFLJLZGSOJAILWDCCNZFHSQHIVZKUC
GOZAAFJEVHOCZVSNHYEHBERLINPEEKRRRHBUXJOPYFF
TXBMMQMOFQS
 - Absolutely incredible odds: 309 million to 1
- Far more impressive than arbitrary ELYOIE sightings

Analyzing Scott's Find...

K4 CT	1 2 3 4	25	42	46	63	84
K4 PT	I	R	N	B	L	E
QUAG Key Position	4	3	9	2	8	7
QUAG Key Letter	F	U	R	M	L	Z

B	E	R	L	I	N
2	7	3	8	4	9
M	Z	U	L	F	R



B	E	R	L	I	N
1	6	2	7	3	8
M	Z	U	L	F	R

Original pattern is +5, +7, +5, +7, +5, ...

New pattern is +5, +5, +5, +5, +5, ...

MUFPZLR or MUFHZLR
work too with +4, +4, +4, ...

Encrypting the "Illusion"

- Scott's Implied Encryption

- Skip-encrypt, using stride of 38, and offset of 77 (degrees on sculpture)

- Not practical to actually do it that way

- Quag III, using "EMUFPHZLRFA"

- Ed's Custom Process (aka Masking Technique)?

- Transposition, spreading BERLIN by amount X

- Quag III, using subset of EMUFPHZLRFA appropriate for X

- Transposition, spreading BERLIN by Y to create a final spread of $X * Y = 38$

- Rotation to create the 77 starting point in K4

The "BERLIN Families"

- I sought to understand matrix methods like columnar transposition, route transposition, Rail-Fence transposition, etc, as well as their inverses.
- I studied Quag III substitutions with varying keyword lengths and observed how they interacted with transpositions
- I found entire families of possible methods and possible paths that would generate the same illusion. I think that this is why we must "solve the technique first and then go for the puzzle"
- I did not just TRY STUFF
- Think Ed couldn't (or wouldn't) do this?
- I think it would be trivial (and fun) for him!



Favorite Family Member Step 1

- Rotate K4 one character to the right...

R O B K R U O X O G H U L B S O L I F B
B W F L R V Q Q P R N G K S S O T W T
Q S J Q S S E K Z Z W A T J K L U D I A
W I N F B N Y P V T T M Z F P K W G D
K Z X T J C D I G K U H U A U E K C A

- Is this the reason for the "?" in front of K4?

Favorite Family Member Step 2

- Input (from last step):

- R O B K R U O X O G...

- Perform a columnar transposition decrypt using 5 columns entered using key (1, 3, 5, 2, 4). See right->

- Output (read off by rows):

- R Q K B W O S Z W I...

- Is this the reason for the conspicuous "point five" found in the coordinates at the end of K2?

1	3	5	2	4
R	Q	K	B	W
O	S	Z	W	I
B	J	X	F	N
K	Q	T	L	F
R	S	J	R	B
U	S	C	V	N
O	E	D	Q	Y
X	K	I	Q	P
O	Z	G	P	V
G	Z	K	R	T
H	W	U	N	T
U	A	H	G	M
L	T	U	K	Z
B	J	A	S	F
S	K	U	S	P
O	L	E	O	K
L	U	K	T	W
I	D	C	W	G
F	I	A	T	D
B	A			

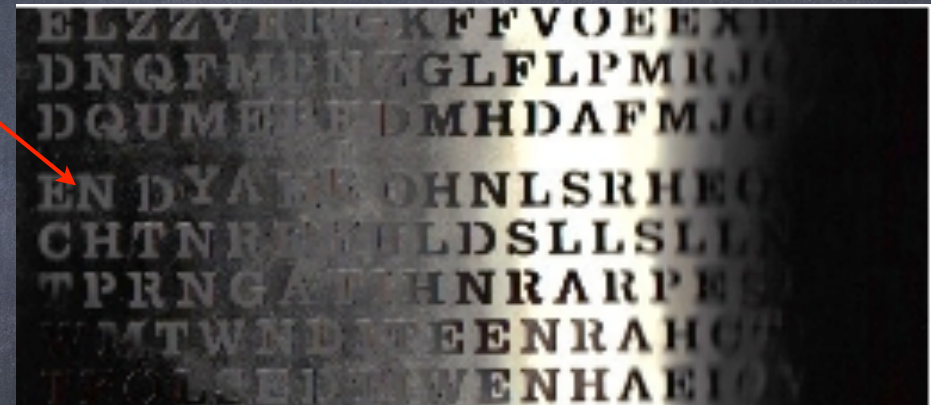
Favorite Family Member Step 3

- Input (from last step): R Q K B W O S Z W I...
- Perform a Quagmire III decryption using the 9 central characters of "EMUF" which are MUFPHZLRF

- Is this the reason for the (possibly) 9 displaced characters of K3 CT?

- It's not just YAR!

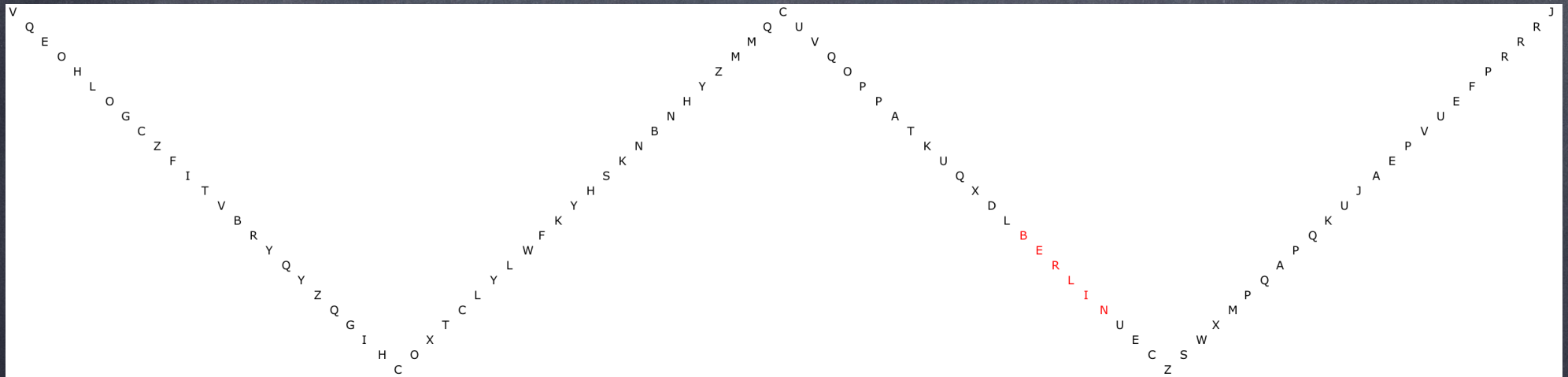
- Steganography!



- Output:
CZHOCSEXWGTUXQCNMZLIPYYLQQQLRAYWEPRFBQBKLKV
YDUTHXJISQAFKUEZKNPCBTVGNAUOHPELYPFHZOPOMQRE
MVRQQURVCJ

Favorite Family Member Step 4

- Input (from last step): C Z H O C S I X E W...
- Perform a bottom's-up Rail Fence decryption using 25 rails. With 97 characters, we get a perfect W...



- Is this the reason for referring to William Webster as "WW"? (Makes you think of Rail-Fence.)
- "BERLIN" is exactly where it needs to be

Favorite Family Member Result

- VQEOHLOGCZFITVBRYQYZQGIHCOXTCLYLWFKYHSKNBNHY
ZMMQCUVQOPPATKUQXDL **BERLIN**UECZSWXMPQAPQKUJA
ERVUEFPRRRJ
- Not correct, but it may be that this method, or some similar family member, is what Ed intended
 - But where is "CLOCK" (or any other words)?!?!?
 - Did Jim add something more to stump Ed (and us)?
 - Berlin Clock encryption?
 - More transposition?
- New Model...



K4 PT -> Jim's Custom Process -> Ed's Custom Process -> K4 CT

Seeking Jim's Custom Process

- One of the family members contains both "BERLIN" and "KRYPTOS", but only if the key has two letters modified in an unmotivated way. Very intriguing
- Performing an additional Quagmire III decryption on the result from the last chart using a key consisting of all "F"s produces the letters "BULB". Major relevance to the history of the "Berlin Clock"
- Performing an additional Quagmire III decryption on the result from the last chart using a key consisting of all "L"s produces the letters "GOLD". Wishful thinking?!?!?
- Using a slightly different process "HANDS" (all Vs) and "HOUR" (all Ls) also appear. More clock relevance

Seeking Jim's Custom Process

- 👁 Another curiosity, using the results of the previous step:

V Q E O H L O G C Z...

- Perform a "Skip-decrypt" with stride of 79 (or -18)

- Perform a Quagmire III decryption using the key "XQQQQQQQQQQV"

👁️ Note arrow on line 2

- Also note 4 Qs, X, and V

[illegible]

- 👁 Output:

XGJBENLZYPFPTSZNESLXIJBKVTQICURWYCPDSKI
VHXDOYLESBOOKZQWVWIBDJBGFOKCWOTIAWKMAAXRH
UQZPKWULOCQB

Seeking Jim's Custom Process

- Result from previous page:
XGJBENLZYPFPTSZNESLXIJWTICBKVKTQICCURWYCPDSKI
VHXDOYLESBOOKZQWVWIBDJBGFOKCWOTIAWKMAAXRH
UQZPKWULOCQB
- Ex-OSS and CIA agent David Doyle wrote these books...
 - An Accurate Watch (William Morrow and Co, 1990)
 - This may have "Berlin Clock" relevance
 - True Man and Traitors: From the OSS to the CIA, My Life in the Shadows (2000)
 - Book not Kryptos relevant, but Doyle might be
- Interesting, but could be a coincidence

Conclusion

- OK, so my efforts to extend Scott Perry's find have obviously not given us the answer
- But we now have a family of practical methods for obtaining BERLIN in the right location that are based on Scott's observation and which all give different results
- Consider this model again...

K4 PT → Jim's Custom Process → Ed's Custom Process → K4 CT

- If Jim really added custom encryption on top of Ed's design, trying to stump Ed, then it is highly unlikely that we will be able to solve K4 without receiving acknowledgement from Ed that they have solved his part of the puzzle

Plea to Ed...



- If the following 3 statements are true:
 - Jim changed the plaintext,
 - Jim altered the encryption by adding another step, and
 - Somebody demonstrates understanding of your intended method,
- Then your obligation to keep the secret has ended. We would be in the same boat and on the same side
- If this is the case, please give a head nod to indicate that there is merit on this path
- Note: at the workshop, Ed laughed at this obvious attempt to "turn" him